MIL-R-5532E (WP)
1 FEBRUARY 1965
SUPERSEDING
MIL-R-5532D (WEP)
15 FEBRUARY 1962

MILITARY SPECIFICATION

RADAR RECOGNITION SET AN/APX-7(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification is an airborne Interrogator-Responsor for use in conjunction with a system of electronic identification and recognition. It shall interrogate Radar Identification Set AN/APX-6B, receive the resulting AN/APX-6B replies and develop video outputs suitable for display on the indicator(s) associated with AN/APX-20A, AN/APS-20B, AN/APS-20C, AN/APS-44 and AN/APS-80 Radar Sets hereinafter referred to as the "local radar".

* 1.2 Classification - The equipment covered by this specification shall be of one type and shall consist of the following items:

Item	Type Designation	Appl. Para.
Coder-Synchronizer	KY-84B/APX-7	3.4.1
Receiver-Transmitter, Radar	RT-261B/APX-7	3.4.2
Control, Radar Set	C-1040A/APX-7	3.4.3
Mounting (for Coder-Synchronizer)	MT-1078/APX-7	3.4.1.2
Mounting (for Receiver-Transmitter)	MT-1208/APX-7	3.4.2.2
Filter, Bandpass	F-943/APX-7	3.4.4

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids except the dated document indicated below, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:09Z
Check the source to verify that this is the current version before use.

MIL-E-4682	Electron Tubes and Transistors. Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft. General Specification for
MIL-T-5422	Testing, Environmental: Aircraft Electronic Equipment
MIL-I-6181B (29 May 1953)	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment (See 2.2(2))
MIL-C-6781	Control Panel: Aircraft Equipment. Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-R-7893	Radar Identification Set AN/APX-6B
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts. Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment. General Specification for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

*2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Source: https://assist.dla.mil -- Downloaded: 202610-17T05:09Z
Check the source to verify that this is the current version before use.

(2) Specification MIL-I-6181B is available only from Bureau of Naval Weapons, Washington 25, D. C. Attention: [illegible]-34.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.7)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

*3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, as outlined in Specification MIL-E-4682.

3.2 Design and Construction -

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 94 pounds.

3.2.3 Operational Stability - The equipment shall operate with optimum performance for 30 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.5 Operating Life -

3.2.4.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 250 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.4.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.4.3 Mean Operating Life Between Failures - The equipment shall have a minimum of 250 hours of mean operating life between failures when tested and accepted as outlined under the requirements of 4.4.2.

Source: https://assist.dla.mil -- Downloaded: 202610-17T05:09Z
Check the source to verify that this is the current version before use.

3.2.5 Cabling and Connections -

3.2.5.1 Cables and Connectors - This equipment shall provide the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.5.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5086. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.6 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for the equipment shall be in accordance with Specification MIL-N-18307.

3.2.7 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181B with the following exceptions:

(1) Figure 16 - This requirement is waived to the extent that noise level limits below 0.21 mc may be exceeded by not more than 50%.

(2) Figure 17 - The broad band radiated interference limits at frequencies of 38, 115, 235, 345, and 690 mc shall be increased to 1.54, 2.8, 2.1, 11.06, and 6.16 microvolts per kc respectively. In addition, it is permissible for any other 2 frequencies encompassed by Figure 17 to exceed the limits by 12 db.

3.2.7.1 Modes of Operation - The interference tests shall be made with the equipment operating in each, all, and any combination of the three modes of interrogation with trigger pulse rates of 300 cps.

3.2.7.2 Cable Lengths - The interference tests shall be made with the interconnecting cable lengths of 20 feet.

3.2.7.3 Conducted Interference Tests - In order to simplify the test procedure, conducted interference tests on specific interconnecting cables may be eliminated provided that these cables do not connect to other electronic systems in the airplane such as radar equipments or primary power supply sources.

* 3.2.7.4 Undesired Antenna Radiation - The power developed at the "Antenna" jack shall not exceed 80 db + 10 log PO below the transmitted signal except at the 1/3 and 9th harmonic of transmitted frequency, which shall be 3000 uv maximum at 343 mc and 25,000 uv at 9270 mc. (PO shall be construed to be the level, in peak watts, of the fundamental transmitted power.)

3.2.7.5 RF Rejection - RF Rejection shall be provided to reduce the sensitivity to at least 40 db below normal sensitivity for all signals in the following ranges: .1 mc through 10 kc. (The use of external filter to meet this requirement is acceptable.)

3.2.7.6 Equipment Protection - The equipment shall not be damaged or rendered inoperative after the application of an RF signal of any amplitude up to two volts (applied at the Antenna jack with a 51 ohm generator calibrated in terms of open circuit voltages) and of any type of modulation.

Source: https://assist.dla.mil -- Downloaded: 202640-17T05:09Z
Check the source to verify that this is the currentversion before use.

3.2.8 Interchangeability - The equipment shall meet the interchangeability requirements as specified in Specification MIL-E-5400.

3.2.9 Maintenance Provisions and Field Testing - Provisions for maintenance of the equipment shall be in accordance with Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.10 Voltage Protection - The equipment shall not be damaged by voltages either above or below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.11 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.12 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($30^{o} \pm 3^{o}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V AC and/or 27.5 ± 0.5 V DC

3.2.13 Service Conditions - The equipment shall operate under any of the environmental service conditions or reasonable combination of these conditions in accordance with Specification MIL-E-5400 for Class 1 equipment except as modified herein.

3.2.13.1 Altitude - The equipment shall provide specified performance under atmospheric service conditions at all altitudes from sea level to 50,000 feet except that an automatic reduction of transmitted RF peak pulse power output is permitted at altitudes above 35,000 feet as described in 3.3.6.

3.2.13.2 Shock -

3.2.13.2.1 Equipment - The equipment shall not suffer damage, or subsequently fail to provide specified performance, when it is subjected to 30 impact shocks of 15G's, each shock impulse having a time duration of 11 ± 1 milliseconds. The shock shall be applied with any individual shock-isolators or shock insulating mounting assembly removed. The shock shall be applied in the following directions:

(1) Horizontally, parallel to the major horizontal axis, five shocks in each direction.

(2) Horizontally, at right angles to the major horizontal axis, five shocks in each direction.

(3) Vertically, five shocks in each direction.

3.2.13.2.2 Mounting Base or Individual Isolators - With excursion stops or bumpers in place, and with maximum rated load applied in a normal manner, the mounting base or individual isolators shall withstand the shock tests of 3.2.13.2.1, except that the magnitude of the shocks shall be not less than 30G's, without evidence of failure, and without evidence of the equipment separating from the mount.

Source: https://assist.dla.mil -- Downloaded: 202610-17T05:09Z
Check the source to verify that this is the currentversion before use.

* 3.2.13.3 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves IV and II of Specification MIL-E-5400. Except that vibration with isolators installed shall have a maximum double amplitude of 0.06 inch over the frequency range of 5 cycles to the point of intersection for the 0.06 inch double amplitude line and Curve IV of Specification MIL-E-5400.

3.2.14 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase) 115V, Category "B", 560 V A

(2) DC Power 28V, Category "B", 1 Amp

3.2.15 Warm-Up Time - The warm-up time for the equipment shall not exceed 3 minutes under standard conditions and 5 minutes at extreme service conditions.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Operation - This equipment shall interrogate Radar Identification Set AN/APX-6B as described in Specification MIL-R-7893 and receive the replies transmitted by AN/APX-6B. It shall be capable of operating with and furnishing receiver video outputs to the remote indicators, control indicators, and relay radar systems of AN/APS-20A, AN/APS-20B, AN/APS-20C, AN/APS-44 and AN/APS-80 (local radars) individually, and one at a time.

3.3.2.1 Remote Control - Remote control provisions shall permit operation as follows:

(1) Any number of operators at remote indicators on the same aircraft, and at remote indicators linked through the relay radar systems, may select one or more modes of interrogation, in any combination, without interference to all other operators.

(2) Operators at radar control indicators may select between modes of interrogation without interference to all other operators.

(3) Route video, corresponding to modes which operators have selected, to the video channels with which the operators are associated.

(4) Maintain the equipment in a quiescent state, with respect to each mode, until that mode has been selected for operation.

3.3.3 Frequency Characteristics - The equipment shall operate in the L-band of frequencies as follows:

3.3.3.1 Transmitter - The transmitter frequency shall be adjustable to any frequency from 1010 through 1030 mc.

3.3.3.2 Receiver - The center of the receiver pass band shall be adjustable to any frequency from 1090 through 1110 mc.

Source: https://assist.dla.mil -- Downloaded: 202640-17T05:09Z
Check the source to verify that this is the current version before use.

3.3.3.3 Tuning - Direct crystal control of the receiver and of the transmitter shall be provided. Tuning of the receiver and of the transmitter shall be independent of each other. A change of receiver or transmitter frequency shall be effected by bench adjustment of controls for each, involving the use of adequate test equipment, and replacement of the applicable crystal.

3.3.3.4 Factory Tuning - Unless otherwise specified in the applicable contract, the equipment shall be delivered with the transmitter tuned to 1030 mc and the center of the receiver pass band tuned to 1090 mc. The crystals required for operation at these frequencies shall be furnished and installed by the contractor. Crystals required for operation at other frequencies shall not be furnished except as a part of the preproduction model (refer to 4.2.2.1).

3.3.3.5 Tuning Stability - The operating frequencies of both transmitter and receiver shall be within ± 200 kc of the selected frequencies after switching from standby (no r.f. transmission) to operation on any Mode or combinations of Modes. This requirement shall be met for the 250 hour period specified in 3.2.4.1, under all reasonable combinations and changes of service conditions specified herein, with variations of input trigger rates from 145 to 3000 pps, with repeated switching from standby to transmission on all Modes and combinations thereof, and with varying load mismatches rangeing between the limits of 1.2 to 1.0, and 2.0 to 1.0 VSWR.

3.3.3.6 RF Tube Replacement - The equipment shall permit replacement of tubes associated with the RF circuits without the necessity of altering equipment adjustments other than those associated specifically with the tube to be changed.

* 3.3.4 Antenna Terminal - A common antenna terminal shall be provided for both transmitter and receiver operation, permitting connection to an antenna through the Filter, Bandpass, F-943/APX-7 by means of coaxial cable and connectors specified in 3.2.5.

3.3.5 Coder-Synchronizer Characteristics -

* 3.3.5.1 Trigger Pulse - This circuit shall accept a cosine pulse as produced by the synchronizer of the local radar and shall produce a pulse suitable for triggering and operating the count down cirucit (3.3.5.2) It shall not impose a load on the cosine circuit that will impair its operation. Two input connectors, type UG-290A/U, connected in parallel, shall be AC coupled to the circuit. If required, suitable termination shall be furnished. When not operating with the local radar systems, this circuit shall accept a triggering pulse having the following characteristics:

(1) Polarity - Positive.

(2) Duration - 0.3 to 20.0 microseconds.

(3) Amplitude - Over the range of from 5 to 150 volts.

(4) Rise Time - From 0.5 to 5.0 volts, within 0.2 microsecond.

(5) Repetition Rate - Between 145 and 3000 pulses per second.

* 3.3.5.2 Count-Down Circuit - A counting down circuit shall be provided for the purpose of limiting the duty cycle of the equipment while utilizing high repetition rates for triggering. The circuit shall operate at input trigger rates as high as 3000 pps and shall be adjusted to provide PRF output maximums of 320 (+0, -20) pps. The circuit shall function such that input trigger rates up to 320 (+0, -20) pps shall be passed without counting down and for input trigger rates greater than 320 (+0, -20) pps the output PRF shall be the largest submultiple, of the input trigger rate, between 150 pps and 320 (+0, -20) pps. Count down frequencies are defined as those cosine pulse repetition frequencies at which division of the input (3.3.5.1) pulse repetition frequency, by the count down circuit, changes from one integer to another. This circuit shall be triggered by the trigger pulse (3.3.5.1) and shall produce a single pulse as required for satisfactory operation of the pulse counter (3.3.5.3), the coded interrogator-responser circuits (3.3.5.9) and the delayed trigger (3.3.5.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:09Z
Check the source to verify that this is the current version before use.

3.3.5.3 Pulse Counter - The pulse counter shall accept the output of the count down circuit as specified in 3.3.5.2 and continuously count groups of 2, 4, 8 and 16 pulses, with an accuracy of ± 12.5%, depending upon the setting of the pulse counter control. Coincident with the last pulse in each counted group, the counter shall produce a single pulse with such characteristics as required for operation of the delay gate generator in the manner specified in 3.3.5.4.

* 3.3.5.4 Delay Gate Generator - This circuit shall accept the output pulse of the pulse counter (3.3.5.3) and shall generate a delay gate for each pulse, adjustable in length within the range of 2700 microseconds and 6200 microseconds, measured from the output pulse of the pulse counter. The gate length shall be adjustable by means of the delay gate generator control (3.4.1.6.5.2).

3.3.5.5 Tripple Gate Generator - This circuit shall accept, as a trigger, a pulse coincident with the trailing edge of the delay gate (3.3.5.4) for the purpose of developing enabling gates or Mode 1, Mode 2, and Mode 3 operation successively. There shall be a single "mode" gate for each delay gate, and each of the successive "mode" gates shall start coincident with the trailing edge of the associated delay gate. Each "mode" gate shall be terminated coincident with the starting of the next succeeding "mode" gate, so that under no conditions will there be more than a single "mode" gate active at a time. The gates shall cause simultaneous operation of the appropriate Mode determining circuits of the Coder (3.3.5.7) the Coded I-R pulse circuit (3.3.5.9), and the Video Separator (3.3.5.10), for the duration of the gate, for any and all combinations of Modes, when these Modes are selected by any interrogate switch(es).

3.3.5.6 Delayed Trigger - This circuit shall accept the count-down circuit output pulse as specified in 3.3.5.2 and produce a pulse that can be delayed with respect to the trigger pulse by an adjustable amount of 6 to 50 microseconds. This pulse shall have the characteristics necessary to operate the coder (3.3.5.7).

3.3.5.7 Coder - This circuit shall accept the delayed trigger (3.3.5.6) and the proper enabling gates from the triple gate generator (3.3.5.5) and shall deliver such pulses to the transmitter modulator as required to produce the pairs of RF pulses specified in 3.3.5.8.1. The pulses delivered from the coder shall be delayed by a different time interval (3.3.5.8.) for each Mode of operation so that the total delay between a trigger pulse (3.3.5.1) and the second pulse of the RF pair associated therewith, shall be identical on all Modes of interrogation. The coder output shall be adequate for delivery to the Receiver-Transmitter via coaxial cable having a total length of 40 feet. There shall be no emission of RF pulses on any Mode unless that Mode has been selected at one of the control positions.

3.3.5.8 Transmitted Pulse Pairs - The transmitted RF pulse pairs shall have the following characteristics:

3.3.5.8.1 Interrogation Modes - Depending upon the Mode selected, interrogations shall have the following pulse spacings:

Mode	Pulse Spacing (microseconds)
1	3 ± 0.2
2	5 ± 0.2
3	8 ± 0.2

Source: https://assist.dla.mil -- Downloaded: 202610-17T05:09Z
Check the source to verify that this is the current version before use.

3.3.5.8.2 Transmitted Pulse Characteristics - Interrogation pulses shall have the following characteristics:

(1) Duration - Between 0.7 and 1.2 microseconds.

(2) Rise Time - Not greater than 0.2 microsecond.

(3) Delay Time - Not greater than 0.4 microsecond.

(4) Amplitude Variation - Pulse amplitudes shall not vary more than 5% over a period of one second under fixed service conditions.

(5) Amplitude Differential - The amplitude of the second pulse of the interrogation shall be within 10% of the amplitude of the first pulse.

* 3.3.5.9 Coded Interrogator-Responsor (I-R) Pulse - A circuit shall be provided which shall accept the output of the count-down circuit (3.3.5.2) and the enabling mode gate(s) (3.3.5.5) appropriate to the interrogation Mode(s) selected by the interrogate switch(es), and shall produce a delayed pulse peculiar to the Mode selected which can be mixed with the data transmission information generated in the synchronizer of the local radar. The indicators associated with the synchronizer video channel (3.3.5.11) will utilize the coded I-R pulse for separation of the modes of IFF video associated therewith. The delayed pulse shall be available at two parallel output connectors, type UG-290A/U, DC connected, and terminated by a 100 ± 5% ohm resistor, for connection to the local radar synchronizer. The delay, measured from the effective cosine pulse (3.3.5.1) which operates this circuit to the coded I-R pulse, shall be 15 ± 0.5 microseconds on Mode 1, 20 ± 0.5 microseconds on Mode 2, 25 ± 0.5 microseconds on Mode 3. The pulse shall have the following characteristics:

(1) Polarity - Positive.

(2) Amplitude - 10 or more volts (measured across the output connector in parallel with 500 micromicrofarads).

(3) Pulse Length - 0.5 to 1.5 microseconds.

(4) Rise Time - From 0.5 to 5.0 volts, within 0.2 microseconds.

3.3.5.10 Video Separator - The video separator shall accept the IFF video from the receiver as specified in 3.3.7.7 and the outputs of the triple gate generator (3.3.5.5). The triple gates shall operate the separator so that the IFF video for the three Modes shall be electronically channelled into the appropriate video output channels (3.3.5.12 and 3.3.5.13) when the selection of interrogation Modes is made by interrogate switches associated with those channels.

* 3.3.5.11 Synchronizer Video Channel - A connector shall be provided to permit selection of all combinations of the Modes of interrogator - responsor RF transmission and operation of the associated video time sharing circuits of the local radar synchronizer. The connector will be used with the Relay Mode Switches (3.4.3.6.4) of the Radar Set Control, or alternatively, with a radio control link. The IFF video corresponding to the selected Modes of interrogation shall be available at a single UG-290A/U connector for connection to the local radar synchronizer (3.4.1.5.1). This connector shall be DC coupled and terminated by a resistor of 75 ± 20% ohms. A control (3.4.1.6.2) shall be provided for adjusting the video output level from two to six volts (3.3.5.15) measured across the output connector in parallel with 500 micromicrofarads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:09Z
Check the source to verify that this is the current version before use.

* 3.3.5.12 Remote Video Channels - Three separate UG-290A/U connectors shall be provided, one for each Mode of interrogation. Positive video for each Mode shall be available at the appropriate jack (3.4.1.5.1) when that Mode has been selected by any of the interrogate switches. Controls shall be provided for each Mode for the purpose of adjusting the video output levels from 2 to 6 volts, measured across the connector in parallel with a capacity of 500 micromicrofarads. The connectors shall be DC coupled and terminated by 75 ± 20% ohm resistors. A connector shall be provided which will permit selection of RF transmission and video on Mode 1, Mode 2 and Mode 3, individually and in any combination, from controls at the remote indicators.

* 3.3.5.13 Control Indicator Video Channel - This channel shall contain the radar video from the local radar systems and positive video for IFF replies corresponding to selected Modes of interrogation. The "MODE" switch (3.4.3.6.2) on the Radar Set Control shall permit selection between Modes 1, 2 and 3 for RF transmission and display of replies when the "CHALLENGE" switch (3.4.3.6.2) is operated. The "VIDEO" switch (3.4.3.6.1) shall make video available at a single DC coupled UG-290A/U connector, terminated by a resistor of 75 ± 20% ohms, as follows:

(1) Radar video only.

(2) IFF video only. Modes 1, 2 or 3 as selected by the "MODE" switch.

(3) Radar video and IFF video of the Mode selected by "MODE" switch.

A control shall be provided for adjusting the video output level between 2 and 6 volts measured across the output connector in parallel with 500 micromicrofarads. The adjusted value shall be the same for all positions of the "VIDEO" switch. The "RADAR VIDEO" connector shall be a single AC coupled UG-290A/U connector. Terminations shall be made as necessary to accept radar video.

3.3.5.14 Video Switch - A video switch shall be provided in the control indicator video channel (3.3.5.1.3) to provide radar video to the "CONTROL IND. VIDEO" terminal whenever the AC power to the Coder Synchronizer is turned off or fails.

* 3.3.5.15 Video Response - The video channels referred to in 3.3.5.11, 3.3.5.12 and 3.3.5.13 shall have the same video response as stated in 3.3.7.7.1, except that the amplifiers shall not degrade the rise and fall times of the pulses to more than 0.15 microsecond. These characteristics shall be measured at the Coder-Synchronizer video output terminals.

3.3.5.16 Video Levels - All outputs of the video channels (3.3.5.11, 3.3.5.12 and 3.3.5.13) shall limit at 6 volts when the receiver output is 2 volts of video, and the output controls are adjusted for 6 volts of video. When the output controls are adjusted to levels other than 6 volts limiting shall occur at those levels. The manual gain control shall be adjusted to provide a ratio of 5 to 1 between limited video and peak noise at the receiver output in all cases. Spurious signals and deterioration of display caused by switching or line transients shall not be present in the video output.

* 3.3.5.17 Video Limiting Stability - The video output limiting levels, as adjusted by the output controls, shall maintain adjusted values of within ± 5% under all reasonable combinations and changes of service conditions specified herein. There shall be less than 0.1 volt peak to peak ripple voltage appearing on any portion of the video output due to power supplies, at all frequencies.

3.3.5.18 Remote Operation - The Mode determining circuits shall be capable of operation from remote points by the grounding of three leads in the Coder-Synchronizer, one lead for each Mode. Routing of selected video in the Coder-Synchronizer by remote control shall be effected by the grounding of three leads in the Coder-Synchronizer, one lead for each Mode. The number of groups of three leads shall be kept to the minimum consistent with the requirements of this specification. Transmission of RF pulse pairs on any Mode shall not occur unless that Mode has been selected at one of the control points. Selection of Modes through one remote channel shall not interfere with the available selections through all other remote channels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:09Z
Check the source to verify that this is the current version before use.

3.3.6 Transmitter Characteristics -

3.3.6.1 Initial Trigger - Pulses as required to produce transmitted pairs of RF pulses specified in 3.3.5.8 shall be furnished to the transmitter modulator by the coder (3.3.5.7).

3.3.6.2 Transmitter Power - The peak pulse power output of the transmitter when passed through a Filter F-943/APX-7 and measured at the Filter output jack terminated by a mismatched load having a voltage standing wave ratio of 1.75 ± 0.05 to 1.0 shall not be less than 2000 watts when any Mode or combination of Modes is transmitted at rates of from 145 to 320 interrogations (pulse pairs) per second, and while operated up to 35,000 + 10% - 15% feet. An automatic pressure-sensitive device shall be incorporated in the equipment which will automatically reduce the power output at altitudes not less than 35,000 feet and permit operation up to 50,000 feet. The minimum power output at altitudes above 35,000 feet and up to at least 50,000 feet shall be 1000 watts.

. 3.3.6.3 Interrogations Characteristics - One complete interrogation shall consist of a pair of RF pulses having the spacings and characteristics specified in 3.3.5.8.1 and 3.3.5.8.2.

3.3.6.4 Transmitter Delay - Transmitter delay measured from the minimum voltage point on the trigger pulse which causes full triggering of the transmitter, to the 50% amplitude point of the first pulse of the pair of output RF pulses shall not exceed 1.0 microsecond.

* 3.3.6.5 Jitter - The jitter of RF pulses shall not be more than 0.10 microsecond with respect to the corresponding point on the leading edge of the initial and effective trigger (3.3.5.1) pulse, except that the jitter shall be limited to 3.0 microseconds at input pulse repetition frequencies (see paragraph 3.3.5.1(5)) within a 20 cps segment located, but not necessarily centered, on any nine (9) of the count-down frequencies defined in paragraph 3.3.5.2.

* 3.3.6.6 Suppression Pulse - The equipment shall provide for the generation and delivery of a Suppression Pulse for the purpose of desensitizing Radar Identification Set AN/APX-6B. A Suppression Pulse shall be generated for each interrogation transmitted and shall precede the first RF interrogation pulse by at least 0.1 microsecond. A UG-290A/U connector shall be provided on the front panel and shall be labelled "SUP". This terminal shall be DC coupled. A screw-driver operated control, equipped with a locking device to permit adjustment of the Suppression Pulse duration, shall be supplied in the Receiver-Transmitter.

* 3.3.6.6.1 Pulse Characteristics - The Suppression Pulse shall have the following characteristics when applied to a load consisting of 2200 ± 5% ohms resistance in parallel with 500 micromicrofarads capacitance:

(1) Amplitude - 15 to 25 volts.

(2) Polarity - Positive

(3) Duration (Measured at 10 volt amplitude) - 150 to 400 microseconds, adjustable.

(4) Rise Time - Not less than 20 volts per microsecond.

(5) Decay Time (Measured from 15 to 3 volts) - Not to exceed 10 microseconds (jitter on the trailing edge of ± 5.0 microseconds is permissable.

* 3.3.6.7 RF Probe - A directional RF probe shall be provided in the line between the transmitter final amplifier and the "ANTENNA" jack. The probe shall terminate in a UG-414A/U adapter and shall produce a standing wave ratio of not more than 2 db in a 51 ohm line connected to it. The probe shall not cause a standing wave ratio of more than 0.5 db in the final amplifier-to-antenna line. When the "ANTENNA" jack is terminated with a 51 ohm resistive load, the power appearing at the probe shall be 27.5 ± 1.0 db less than that flowing toward the load, and shall be at least 45.5 db less than that flowing away from the load. The attenuation of the probe in db at the HIGH END of the transmitter frequency range shall be marked on the front panel adjacent to the UG-414A/U adapter. This attenuation marking shall be accurate to ± 0.5 db. A curve, table, or formula, which

Source: https://assist.dla.mil -- Downloaded: 202610-17T05:09Z
Check the source to verify that this is the currentversion before use.

will permit determination of the attenuation to within ± 0.5 db at all other transmitter frequencies shall be furnished in the applicable handbook.

3.3.6.8 High Voltage Delay - Automatic provisions shall be incorporated within the equipment to prevent the modulator from applying pulses to the transmitter, after the power relay has been energized, until the tube cathodes are heated. The design of this automatic warm-up provision shall be such as to function in the event of a failure of the primary AC power supply and shall be based on the assumption that the AC power failure will not normally be accompanied by a similar DC power failure; however, operation of this delay shall be independent of DC.

3.3.6.9 Power Monitor - A power monitor shall be incorporated which will connect a pin of the control connector (J108-D) through a low impedance to ground in the event the peak transmitter power falls below a preset reference. This connector pin will be used to supply a ground return for an external indicator lamp which will serve as a warning that either the transmitter is inoperative or operating with reduced power. The ground return shall consist of a switching transistor which provides less than 300 ohms impedance for a 20 ma load when "on" and less than 800ua leakage for 28 VDC alarm supply when "off". The maximum alarm load shall be 20 ma and the maximum alarm supply shall be 29 VDC. The alarm actuation point shall be adjustable by a control located on the power monitor module through a minimum range of +30 to +35 db W. The actuation point shall be stable to ± 2 db over service conditions (goal ± 1 db). A spring return switch shall be located on the front panel which will operate the modulator altitude switch and thus reduce the transmitter power. This switch will be used for calibration and/or self test of the power monitor.

3.3.6.10 Twin Pulse Generator - A twin pulse generator shall be incorporated which will modulate the transmitter with mode 1 interrogations when a single trigger is applied to a type BNC front panel jack. This jack shall be labeled "AUX TRIG" and located under the protective flap. This auxiliary trigger shall have the following characteristics:

Amplitude:	20 ± 5v into 100 ohms (positive)
Width:	1 + 2. -0 microsecond
Rise Time:	0.1 microsecond
Fall Time:	0.2 microsecond
PRF:	160 - 320

The circuitry shall be designed so that the transmitter may be triggered from either the normal TRIG jack or the AUX TRIG jack without mechanical switching. This auxiliary trigger shall also trigger Receiver Gate, GTC, and Suppression unless these functions have been otherwise disabled by the applicable circuitry.

3.3.7 Receiver Characteristics -

3.3.7.1 Transponder Reply - The equipment shall receive replies transmitted by Radar Identification Set AN/APX-6B. The equipment shall also be capable of receiving single or multiple pulse replies having the characteristics as specified herein.

3.3.7.1.1 Pulse Shape - The basic Reply Pulse will have the following characteristics:

(1) Duration - Between 0.4 and 1.3 microseconds.

(2) Rise Time - 0.2 microsecond, or less.

(3) Decay Time - 0.4 microsecond, or less.

(4) Spike Amplitude - Not greater than 40% of the average pulse amplitude.

(5) Amplitude Jitter - 5% or less.

Source: https://assist.dla.mil -- Downloaded: 202640-17T05:09Z
Check the source to verify that this is the current version before use.

* 3.3.7.1.2 Reply Coding - Transponder replies will be coded according to the Mode of interrogation used, as follows:

Mode	No. of Pulses	Pulse Spacing (microseconds)
1	1	----
2	2	16 ± 2.5
3	1	----
Emergency Reply	4	16 ± 2.5
Auxiliary Reply	3 to 28 (duration of 0.45 ± 0.1 microsecond only)	1.45 to 2.9 (trains may be interleaved)

3.3.7.2 Receiver Gain - The maximum receiver gain shall be greater than 106 db (of which 6 db shall be in the video processor and the remainder in the IF amplifier). In order to fulfill this requirement, the receiver shall be capable of producing limited 2 volt video output for an RF input signal at a level of -100 dbv, when the manual gain control is set at maximum. Limiting of the signal may be just discernable in the noise.

* 3.3.7.2.1 Gain Time Control (GTC) - An electronic switch shall be provided, which, when not being triggered shall hold the receiver desensitized to at least 75 db less than normal sensitivity. (Normal sensitivity is the signal required to produce "just limiting" of the video output. The "normal sensitivity" may be adjusted for any level between -60 and -100 dbv.) When the electronic switch is triggered by a pulse coincident with the first RF pulse of an interrogation, the sensitivity shall rise from the desensitized level to an adjustable initial GTC level. From this level the receiver sensitivity shall be increased at the rate of 6 ± 2 db/octave. The time at which the receiver sensitivity is 3 db below normal sensitivity shall be termed the GTC period. This GTC period shall be adjustable from 400 to 2400 microseconds by a chassis-mounted potentiometer. The interval for which the receiver gain is increased at 6 ± 2 db/octave shall be at least the final 5 octaves, or at least include the interval between 50 microseconds after triggering and the GTC period. The receiver shall remain sensitized at normal sensitivity for the receiver gate period, after which it shall return to a desensitized condition until the next effective trigger causes a repetition of the cycle.

3.3.7.2.2 Recovery Time - With the GTC circuit switched ON, the receiver shall recover as specified in 3.3.7.2.1. When the GTC circuit is disabled, the receiver shall recover to within 20 db of normal sensitivity in not more than 15 microseconds, measured from the trigger pulse, and within 3 db of normal sensitivity in not more than 25 microseconds, measured from the trigger pulse.

3.3.7.2.3 Receiver Gate - The receiver shall incorporate a gate which, when actuated by the trigger pulse, will limit the period in which it can develop a video output. The length of this gate shall be adjustable throughout the minimum range of 2500 to 6200 microseconds, measured from the trigger pulse. This gate period shall have a trailing edge jitter of less than ± 0.1% and shall be stable to within ± 1.5% over the entire range of environmental conditions. The recovery time of the receiver gate shall be sufficiently short as to allow the following receiver gate interval to start 60 microseconds after the conclusion of the receiver gate period.

3.3.7.2.3.1 Receiver Gate Operation - With the GTC circuit disabled, the receiver shall provide normal sensitivity following the specified recovery time for a period established by the receiver gate. At the end of this time, the receiver shall be desensitized to at least 75 db less than normal sensitivity as rapidly as practicable but within 35 microseconds, as measured between the 3 and 75 db points, until the next trigger pulse causes a repetition of the operating cycle. With the GTC circuit enabled, the receiver shall provide reduced sensitivity as specified in the preceding paragraphs followed by normal sensitivity for the remainder of the period established by the gate after which the receiver shall be desensitized to at least 75 db below normal sensitivity until the next external trigger causes a repetition of the operation cycle. By operation of the front panel GTC and receiver gate switches, it shall be possible to independently select any one of the following conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:09Z
Check the source to verify that this is the current version before use.

a. GTC only

b. Receiver gate only

c. Both GTC and receiver gate

d. A receiver on condition (Gain determined by manual gain control)

* 3.3.7.3 Noise Figure - The noise figured of the receiver shall be less than 13 db for operation between -54°C and +25°C and at any ambient temperature between -25°C and the maximum operating temperature and at any altitude between ground and the maximum specified operating altitude, the noise figure shall be less than 14.5 db or (whichever is less):

$$N.F. \leq 13 \text{ db} + .06A \text{ db} + .1 (T-25) \text{ db}$$

Where: N.F. = Noise Figure
A = Test altitude in thousands of feet
T = Test temperature in degrees centigrade

* 3.3.7.4 Receiver Overload - The receiver shall not overload when a series of RF pulse trains are delivered, at the "ANTENNA" jack with the following characteristics:

(1) Number of Pulses - 28

(2) Pulse Duration - 0.45 ± 0.1 microsecond

(3) Rise and Fall Time - 0.1 microsecond, or less

(4) Pulse spacing - Not less than 1.0 microsecond, nor more than 3.0 microseconds

(5) Repetition Rate - 600 trains per second, or more. However, the duty cycle shall not exceed 25%.

(6) Frequency Range - 1090 through 1110 mc

(7) Level - 20 db below one volt

* 3.3.7.5 Selectivity - The receiver shall have a frequency response which closely approximates a Gaussian or "bell-shaped" response. The overall bandwidth shall be greater than 7 mc at 3 db below maximum, greater than 9 mc at 6 db below maximum, not greater than 22 mc at 40 db below maximum, and not greater than 27 mc at 60 db below maximum. The sensitivity to the image frequency shall be greater than 50 db below sensitivity at the selected channel. The sensitivity to all other frequencies shall be greater than 60 db below the sensitivity to the selected channel. Signals ± 3 mc from center frequency shall not reduce sensitivity by more than 3 db.

3.3.7.6 Intermediate Frequency Amplifier -

3.3.7.6.1 Frequency - The receiver shall employ an IF amplifier having a frequency of 59.5 ± 0.5 mc at the arithmetic mean between the two frequencies where the response is 6 db below maximum.

3.3.7.6.2 IF Attenuation - The IF attenuation measured at the "ANTENNA" jack shall not be less than 70 db.

3.3.7.6.3 IF Overload - The overload characteristics of the IF amplifier shall be consistent with the receiver overload characteristic (3.3.7.4).

Source: https://assist.dla.mil -- Downloaded: 202610-17T05:09Z
Check the source to verify that this is the current version before use.

3.3.7.6.4 Manual IF Gain Control - A knob of the Radar Set Control (3.4.3.6.5) shall control the gain of the IF amplifier. When this control is set at maximum, the peak noise output at the receiver video terminal shall be saturated. The relation between knob angular displacement and output attenuation in db shall be approximately linear. The control shall be capable of inserting at least 55 db of attenuation. There shall be no more than 36 db gain in the stages ahead of the controlled stage. The time constant of the gain control circuit shall be less than 0.1 second. The gain control circuits shall be filtered against high frequency transmission.

3.3.7.6.5 Gain Stability - The amplitude of the video output shall not vary more than ± 5% over a 5 hour period under all reasonable combinations and changes of service conditions with an input signal 6 db above Normal Sensitivity.

* 3.3.7.7 Video Output - Two UG-290A/U video output terminals shall be connected in parallel on the front panel of the receiver. They are to be designated "VIDEO". These terminals shall be DC coupled. The pulse amplitude level shall be adjustable by a pre-set control (3.3.7.7.2) on the front panel to provide positive 2 volts when the terminal is loaded by a 500 micromicrofarad capacitor. The terminal shall be terminated by a resistor of 75 ± 5% ohms. There shall be less than 0.1-volt ripple voltage at all frequencies measured across the termination resistor from the power supplies.

* 3.3.7.7.1 Video Fidelity - The fidelity of the receiver shall be checked by applying RF pulses to the antenna jack and observing the video output of the receiver. The RF pulses shall be the standard output from an AN/UPM-99 or equivalent test set and within ± 2 mc of the receiver center frequency. The receiver shall be capable of accepting RF pulse trains within the range of -20 dbv to -100 dbv without overload. Any combination of pulse duration, pulse spacing, and repetition rate may be used provided that the duty cycle does not exceed 25%.

3.3.7.7.1.1 Rise Time - The rise time of all output pulses with inputs greater than 3 db above normal sensitivity shall be less than 0.08 microseconds.

3.3.7.7.1.2 Fall Time - The output fall time for input pulses greater than 3 db above normal sensitivity shall be less than 0.10 microsecond.

* 3.3.7.7.1.3 Video Limiter - A control located on the front panel shall be provided to limit the video level at the receiver output terminals under load conditions specified above at 2 volts. This level shall be independent of gain setting and shall be stable within ± 5% over all environmental conditions. This level shall be measured by observing the signals on a calibrated oscilloscope.

3.3.7.7.2 Dynamic Range and Pulse Fidelity - The receiver shall preserve the input pulse train widths and spacings for all RF signal strengths from just great enough to produce limiting to 40 db above that level.

3.3.7.7.2.1 Pulse Train Fidelity - The receiver performance shall be tested with an AN/UPM-99 or equivalent test set generating a code 7777 train. The video output shall not shift the spacing or width of any pulses more than ± 0.05 microsecond over a 40 db dynamic range. A shift of 0.08 microsecond in one direction will be acceptable provided the shift in the other direction is less than 0.02 microsecond. No spurious outputs shall occur in this range.

3.3.7.7.2.2 Interleaved Pulse Train Fidelity - The performance test of 3.3.7.7.2.1 shall also be performed with two mode 2 pulse trains interleaved to within 0.7 microsecond at the leading edges and separated in level by at least 40 db. This shall apply to trains of either like or unlike code. The tolerances of 3.3.7.7.2.1 shall apply. These requirements shall also apply for a dynamic range of 40 db or greater provided the closest spacing between pulses is not less than 0.85 microsecond.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:09Z
Check the source to verify that this is the current version before use.

3.3.7.7.2.3 Operation with Degraded Pulse Shapes - A pulse-for-pulse IAGC system shall be incorporated in the IF amplifier and Video Processor to enable the receiver to meet the requirements of 3.3.7.7.1 through 3.3.7.7.2.2. It shall also meet these requirements over a dynamic range of 30 to 35 db with RF input pulses having a rise time as great as 0.15 microsecond and a fall time as great as 0.35 microsecond.

3.3.7.8 Receiver Delay - The delay between an RF pulse applied to the antenna jack and the corresponding pulse at the video output jack shall not exceed 0.6 microsecond. The delay shall be measured between the 50% amplitude points on the leading edges of the pulses with the manual gain control set for "just limiting" at any input level between -74 dbv and -94 dbv.

3.3.7.9 Anti-jamming - The receiver shall be capable of operating in the presence of jamming as follows:

3.3.7.9.1 CW - A pulse signal having a voltage amplitude of not more than 84 db below one volt, shall produce useful video output in the presence of CW jamming at not less than 66 db below one volt.

3.3.7.9.2 MCW - In the presence of CW modulated 30% at 1000 cps at a level of not less than 72 db below one volt, the video output shall not contain more than one volt peak-to-peak envelope of 1000 cps and its harmonics. The manual "GAIN" control shall be set for 0.25 rms volts of noise as measured at the second detector.

3.3.7.9.3 Circuitry - The use of circuits which impair equipment operation in the presence of railing and FM jamming shall be avoided whenever practicable. The anti-jamming circuits used shall not appreciably impair the normal operation of the receiver in the absence of jamming.

* 3.3.7.10 Oscilloscope Probes - Provisions shall be incorporated for monitoring the 2nd detector output and the transmitter pulse shape with an external oscilloscope connected to type BNC jacks located under the protective flap on the front panel. These jacks shall be capable of driving at least 30 feet of RG-62/U or similar coaxial cable terminated in 93 ohms. A transmitter pulse amplitude of 8 volts peak shall be considered nominal.

3.3.5.11 IFPM Enabling - Circuitry shall be incorporated which will disable the functions of Receiver Gate, GTC, and Suppression when a pin on the power connector (J107-G) is grounded by an external switch. This circuitry shall be independent on the front panel Receiver Gate ON-OFF and GTC ON-OFF switches.

3.4 Detailed Requirements -

* 3.4.1 Coder-Synchronizer KY-84B/APX-7 - The Coder Synchronizer shall meet the following requirements:

3.4.1.1 Function - This unit shall generate the information as specified in 3.3.5.1 through 3.3.5.18.

* 3.4.1.2 Form Factor and Mounting - Case and mounting base shall be in accordance with the following:

(1) The size of the case shall not exceed 19-11/16 by 11-1/8 by 8-13/16 inches. It shall be mounted by Mounting MT-1078/APX-7.

(2) The size of the Mounting MT-1078/APX-7 shall not exceed 19-7/16 by 11-1/8 by 2-7/8 inches. The weight shall not exceed 2.5 pounds.

Source: https://assist.dla.mil -- Downloaded: 202640-17T05:09Z
Check the source to verify that this is the current version before use.

3.4.1.3 Weight - The weight of this unit, including tubes, crystals, connectors and mounting base shall not exceed 40 pounds.

3.4.1.4 Contents - This unit shall contain all the circuits necessary to generate the information as specified in 3.3.5.1 through 3.3.5.18. It may include all or part of the required power supplies.

3.4.1.5 Electrical Connections - Electrical connections shall be provided as necessary to connect this unit to associated circuits in accordance with the requirements of this specification. All external electrical connections of this unit utilized in its installation and in pre-flight tests with the equipment installed in aircraft shall be located on the front panel. Power connections shall be in accordance with 3.2.5.3 of this specification.

* 3.4.1.5.1 RF and Video - The following RF and video connections shall be included:

Label	Type and Function
"COS PULSE"	UG-290A/U - Two parallel input jacks for connection to the cosine pulse from the local radar synchronizer as described in 3.3.5.1, and from the test equipment. One jack shall be capped.
"TRIG"	UG-290A/U - One output jack for connection of the first and second triggers to the Receiver-Transmitter as described in 3.3.5.7.
"CODED I-R PULSE"	UG-290A/U - Two parallel output jacks for connection of the coded Interrogator-Responsor pulse to the local radar synchronizer as described in 3.3.5.9. One jack shall be capped.
"I-R VIDEO"	UG-290A/U - One input jack for connection to the Interrogator-Responsor video as described in 3.3.5.10.
"SYNC VIDEO"	UG-290A/U - One output jack for connection of the Interrogator-Responsor video to the local radar synchronizer as described in 3.3.5.11.
"MODE 1" "MODE 2" "MODE 3"	UG-290A/U - Three output jacks for connection of Modes 1, 2 and 3 video outputs as described in 3.3.5.12. These jacks shall be grouped over the label "REMOTE VIDEO".
"RADAR VIDEO"	UG-290A/U - One input jack for connection to the radar video from the local radar synchronizer as described in 3.3.5.13.
"CONTROL IND VIDEO"	UG-290A/U - One output jack for connection of video selected by the "VIDEO" switch (3.4.3.6.1) to the local radar control indicators as described in 3.3.5.13.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:09Z
Check the source to verify that this is the current version before use.

3.4.1.6 Front Panel Controls - The following controls shall be in accordance with 3.2.11, located on the front panel, and labelled as indicated:

3.4.1.6.1 "DELAYED TRIGGER" - This control shall consist of a potentiometer and shall permit adjustment of the delayed trigger as specified in 3.3.5.6.

3.4.1.6.2 "SYNC VIDEO OUTPUT" - This control shall consist of a potentiometer and shall permit adjustment of the "SYNC VIDEO" output as specified in 3.3.5.11.

3.4.1.6.3 "REMOTE VIDEO OUTPUT" - Three separate potentiometers shall permit adjustment of the remote video output levels for each Mode as specified in 3.3.5.12. These controls shall be grouped under the label "REMOTE VIDEO OUTPUT" and the individual controls shall be labelled "MODE 1", "MODE 2", "MODE 3" as appropriate.

3.4.1.6.4 "CONTROL IND VIDEO OUTPUT" - This control shall consist of a potentiometer and shall permit adjustment of the control indicator video output as specified in 3.3.5.13.

3.4.1.6.5 Pulse Counter and Delay Gate Generator Controls -

3.4.1.6.5.1 "PULSE COUNTER" - This control shall be provided to permit operation of the pulse counter as specified in 3.3.5.3. The control shall be calibrated to count pulse groups of "1", "2", "4", "8", and "16" pulses each.

3.4.1.6.5.2 "GATE CONTROL" - This control shall be provided to permit operation of the delay gate generator as specified in 3.3.5.4.

3.4.2 Receiver-Transmitter RT-261B/APX-7 - The Receiver-Transmitter shall meet the following requirements:

3.4.2.1 Function - This unit shall accept triggers, transmit RF interrogations, receive and detect transponder (AN/APX-6B) replies and shall, through the Coder-Synchronizer, present these replies as video signals for display on associated equipment.

* 3.4.2.2 Form Factor and Mounting - Case and mounting base shall be in accordance with the following:

(1) The size of the case shall not exceed 19-3/4 by 10-3/4 by 8 inches. It shall be mounted by Mounting MT-1208/APX-7.

(2) The size of the Mounting MT-1208/APX-7 shall not exceed 19-7/16 by 11-1/8 by 2-7/8 inches and the weight shall not exceed 2.5 pounds.

3.4.2.3 Weight - The weight of this unit, including tubes, crystals, connectors and mounting base shall not exceed 52 pounds.

3.4.2.4 Contents - This unit shall contain an RF transmit-receive switch or equivalent, receiver, transmitter, video processor, video and suppressor gate, GTC circuit, power monitor, twin pulse generator relays, and may include all or part of the power supplies.

3.4.2.5 Electrical Connections - Electrical connections shall be provided as necessary for interconnection to the other units of the AN/APX-7, for the connection of this equipment to an antenna, for connection of this equipment to Radar Identification Set AN/APX-6B, and for connection of this equipment to primary power supplies, utilizing cables and connectors in accordance with 3.2.5. Provisions for connection to primary power supplies shall be in accordance with 3.2.5.3. All external electrical connections of this unit utilized in its installation and its preflight test (equipment installed in the aircraft) shall be located on the front panel.

Source: https://assist.dla.mil -- Downloaded: 202610-17T05:09Z
Check the source to verify that this is the currentversion before use.

* 3.4.2.5.1 RF and Video - The following RF and Video connections shall be included:

Label	Type and Function
"ANTENNA"	UG-61A/U - One jack for connecting the common antenna line to the transmitter and receiver RF circuits as described in 3.3.4.
"RF PROBE"	UG-414A/U - One jack for terminating the RF probe as described in 3.3.6.7.
"VIDEO"	UG-291C/U - Two parallel output jacks usable interchangeably for connection of the video output as described in 3.3.7.7. The jacks shall be grouped so that the same labels identify both jacks. One jack shall be capped.
"SUP"	UG-290A/U - One output jack for connection of the suppression pulse to the AN/APX-6 equipment as described in 3.3.6.6.
"TRIG"	UG-290A/U - One input jack for connection of the first and second triggers from the Coder-Synchronizer to the Receiver-Transmitter to actuate the Transmitter Modulator, Suppression Pulse, GTC, and Receiver Gate as described in 3.3.6.1, 3.3.6.6, 3.3.7.2.1, and 3.3.7.3, respectively.
"AUX TRIG"	UG-290A/U - One input jack for connection of an auxiliary single trigger to the twin pulse generator as described in 3.3.6.10.
"2ND DETECTOR"	UG-290A/U - One output jack for monitoring the second detector output with an external oscilloscope as described in 3.3.7.10.
"TRANS SHAPE"	UG-290A/U - One output jack for monitoring the transmitter pulse shape with an external oscilloscope as described in 3.3.7.10.

3.4.2.6 Panel Controls, Front - The following controls shall be in accordance with 3.2.11 of this specification, located on the front panel, and labelled as follows:

3.4.2.6.1 "GTC" - This control shall be a 2 position toggle switch equipped for safety-wiring in both the "ON" and "OFF" positions. In the position labelled "ON", the equipment shall perform as specified in 3.3.7.2.1; in the position labelled "OFF", the equipment shall perform as specified in 3.3.7.2.2.

3.4.2.6.2 "RCVR GATE" - This control shall be a 2 position toggle switch equipped for safety-wiring in both the "OFF" and the "ON" positions. In the position labelled "OFF", the receiver shall remain sensitized for maximum periods between the initial external trigger pulses, limited only by the receiver recovery time as specified in 3.3.7.2.2. In the position labelled "ON" the receiver shall remain sensitized for a limited period as determined by the setting of the Receiver Gate Period as specified in 3.4.2.6.5.

3.4.2.6.3 "VIDEO LIMITER" - This control shall consist of a potentiometer and shall permit adjustment of the video output as specified in 3.3.7.7.2.

Source: https://assist.dla.mil -- Downloaded: 2026l0-17T05:09Z
Check the source to verify that this is the current version before use.

3.4.2.6.4 Meter and Selector - For convenient monitoring, a DC meter and a selector-switch shall be grouped on the front panel identified by the label "TEST". This meter and control shall permit measurement of the following:

Selector Switch Label		Measurement Provided
"MIXER XTAL"		Mixer Crystal Current
"MOD HV"		Modulator Screen Voltage
"TRIG IN"		Trigger Current
"XMTR CUR"	"EXC ON"	Overall Modulator Operation
	"EXC OFF"	Transmitter Oscillator and Multiplier Operation Transmitter
		Amplifier Operation

3.4.2.6.5 "TRANS PWR REDUCE" - This control shall be a two-position spring-return toggle switch which shall reduce the transmitter power for calibration of the power monitor as specified in 3.3.6.9.

* 3.4.3 Control, Radar Set C-1040A/APX-7 - The control shall meet the following requirements:

3.4.3.1 General - The design, materials and workmanship used in the construction of this unit shall conform to Specification MIL-C-6781.

3.4.3.2 Function - This unit shall provide for remote control of the AN/APX-7.

* 3.4.3.3 Form Factor and Mounting - The size and mounting provisions of this unit shall be in accordance with the requirements of Specification MIL-C-6781. Maximum dimensions shall not exceed 3-3/8 by 5-3/4 by 4-13/16 inches.

3.4.3.4 Contents - The control shall consist of a plastic lighting panel and built-in red lighting in accordance with Specification MIL-P-7788, a metal mounting plate, the necessary controls, components and associated wiring, an electrical connector, and a cover box.

3.4.3.5 Weight - The weight shall be held to a minimum in accordance with the requirements of Specification MIL-C-6781 and shall not exceed two pounds.

3.4.3.6 Control - The following controls shall be located on the front panel and labelled as indicated:

3.4.3.6.1 "VIDEO" - A 3 position switch shall control the type of video presented to local radar control indicator (3.3.5.13). The following switch positions shall be provided:

Selector Switch Position Label	Operation Provided
"IFF"	AN/APX-7 video only.
"BOTH"	Radar and AN/APX-7 video.
"RADAR"	Radar video only.

Source: https://assist.dla.mil -- Downloaded: 202610-17T05:09Z
Check the source to verify that this is the current version before use.

3.4.3.6.2 "MODE" - A 3 position switch with contol detents labelled "1". "2" and "3" shall permit selection of interrogation Mode and resulting reply to be displayed on the control indicator of the local radar.

3.4.3.6.3 "CHALL" (Challenge) - A 3 position toggle switch shall be connected in series with the common return lead of the "MODE" (3.4.3.6.2) control. The switch positions shall have the following labels and functions:

Switch Position and Label	Operation Provided
Down position. "MOM"	Interrogations are transmitted and replies are displayed while held. Switch is spring loaded to return to "OUT" position when released.
Center position. "OUT"	No interrogations transmitted.
Up position. "ON"	Same as for "MOM" except that switch is not held and must be returned to the "OUT" position manually.

3.4.3.6.4 "RELAY MODE" - Three 2 position toggle switches shall be grouped above the label "RELAY MODE". The down position of these switches shall be marked "OUT". The up position shall be marked "1", "2" and "3". In the up position, these switches shall activate the video time sharing circuits of the local radar synchronizer, cause transmission of interrogations and routing of corresponding video to the "SYNC VIDEO" terminal for the Mode(s) selected. This operation shall be accomplished by grounding leads from the Coder-Synchronizer (3.3.5.11).

3.4.3.6.5 "GAIN" - A control shall permit adjustment of receiver gain as specified in 3.3.7.6.4. The direction of control rotation corresponding to increasing gain shall be clockwise.

3.4.3.6.6 "PWR" (Power) - A 2 position toggle switch shall be provided for controlling the power relay(s). In the down position labelled "OFF", no primary power shall be furnished to the electrical loads of the equipment. In the up position labeled "PWR", the equipment shall be energized.

3.4.3.7 Electrical Connections - One multiple-pin receptacle in accordance with 3.2.5 shall be provided for connecting this unit to the associated circuits of the AN/APX-7, to supply a control lead to the local radar synchronizers, to connect to the lighting power sources, and to permit mode selection and challenge from auxiliary indicator(s).

3.4.3.8 Lighting Circuit - The lamps used for lighting the plastic face of the control shall be connected in parallel, with the +28 V DC input lead connected to a separate contact in the connector. The brilliance of the lamps will be controlled externally, and controls for this purpose shall not be a part of the AN/APX-7.

3.4.3.9 Grounding - Applicable circuits shall be grounded internally, and a ground connection shall be brought out through one of the contacts in the connector.

* 3.4.4 Filter, Bandpass F-943/APX-7 - The filter shall meet the following requirements:

* 3.4.4.1 General - The design, materials, and workmanship used in the construction of the filter shall conform to Specification MIL-E-5400 unless otherwise specified herein.

* 3.4.4.2 Function - The filter shall be installed in the Receiver-Transmitter RT-261B/APX-7 antenna line for the suppression of undesired antenna radiation.

Source: https://assist.dla.mil -- Downloaded: 202640-17T05:09Z
Check the source to verify that this is the current version before use.

* 3.4.4.3 Form Factor and Mounting - The form factor and mounting shall be in accordance with the following:

(1) The size of the filter shall not exceed 5-19/32 inches in length by 3/4 inch in diameter.

(2) The filter shall be mounted external to the Receiver-Transmitter RT-261B/APX-7 with suitable hardware. The hardware shall be furnished by the air frame manufacturer.

* 3.4.4.4 Weight - The weight of the filter shall not exceed 8 ounces.

* 3.4.4.5 Electrical Connections - Electrical connections shall be provided for interconnection between other units of the AN/APX-7, utilizing cables and connectors in accordance with 3.2.5. One type N male connector shall be provided for connection to the antenna line from the RT-261B/APX-7 and one type N female connector shall be provided for connection to the antenna line from the antenna.

* 3.4.4.6 Performance - The receiver-transmitter shall comply with the requirements specified in 3.3. when the filter is externally connected in the RT-261B/APX-7 antenna line. The filter shall perform to the following requirements.

* 3.4.4.6.1 Impedance - The filter shall have a nominal impedance of 50 ohms.

* 3.4.4.6.2 Insertion Loss - The insertion loss shall not exceed 0.5 db when RF pulses in accordance with paragraphs 3.3.5.8.2(1), 3.3.5.8.2(2), and 3.3.5.8.2(3) are applied to the Type N male connector throughout the frequency range of 1010 to 1030 mc or when RF pulses in accordance with paragraphs 3.3.7.4(2) and 3.3.7.4(3) are applied to the Type N female connector throughout the frequency range of 1090 to 1110 mc.

* 3.4.4.6.3 Power Rating - The average power rating shall be more than 15 watts. The peak power rating for a 0.5 microsecond pulse shall be more than 4000 watts for altitudes between 0 and 40,000 feet and shall be more than 2000 watts for altitudes between 40,000 and 50,000 feet.

* 3.4.4.6.4 Voltage Standing Wave Ratio - The voltage standing wave ratio shall not exceed 1.5 to 1.0.

* 3.4.4.6.5 Frequency Response - The filter shall attenuate signals below 680 mc not less than 20 db and shall attenuate signals below 340 mc and above 2,000 mc not less than 40 db. The stop band rejection to 10.0 gc shall be not less than 40 db.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

Source: https://assist.dla.mil -- Downloaded: 202610-17T05:09Z
Check the source to verify that this is the currentversion before use.

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration test shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. This data may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.2.1.1 Special Crystals - The contractor shall furnish as a part of the preproduction model a set of crystals which will permit operation of the transmitter at 1010, 1020, and 1030 mcs and the receiver at 1090, 1100, and 1110 mcs.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181. The sampling and testing of plastic lighting plates shall be in accordance with Specification MIL-P-7788.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is

Source: https://assist.dla.mil -- Downloaded: 20260-17T05:09Z
Check the source to verify that this is the current version before use.

approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Product ion Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Reliability Quality Assurance Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operation Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for

Source: https://assist.dla.mil -- Downloaded: 202640-17T05:09Z
Check the source to verify that this is the currenversion before use.

5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Reliability Quality Assurance Tests - Equipments which have successfully passed the individual tests shall be tested for reliability quality assurance in accordance with Specification MIL-R-23094, Procedure II.

4.4.2.1 Test Level - The Test level shall be Level III (from Table I of Specification MIL-R-23094).

4.4.2.2 ACCEPT-Reject Criteria - Figure 11 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria with MTBF as stated in 3.2.4.3.

4.4.2.3 Detail Test Procedures for Reliability Quality Assurance Tests - The contractor shall prepare detail test procedures for the approval of the procuring activity. The procedures shall include operating time(s), characteristics of the temperature cycle, performance characteristics to be measured, the Failure criteria and maintenance to be allowed during the test periods as well as other requirements necessary for conducting the tests.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (see para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:09Z
Check the source to verify that this is the current version before use.

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

Source: https://assist.dla.mil -- Downloaded: 202610-17T05:09Z
Check the source to verify that this is the currentversion before use.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:09Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

* 6.1 An asterisk preceding a paragraph number denotes a change in that paragraph from the 'D' version of this specification.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages 27.5 ± 0.5 V DC and 115 ± 1.0 V AC, as applicable. These data are to be used in testing the equipment at installation points for compliance to the minimum acceptable standard of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 The parenthese (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.6. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Definitions - The following definitions shall apply.

6.6.1 Pulse Rise Time - Pulse rise time is the time interval in microseconds between the 10 and 90 percent of maximum amplitude points on the leading edge of the pulse.

6.6.2 Pulse Decay Time - Pulse decay time is the time interval in microseconds between the 90 and 10 percent of maximum amplitude points on the trailing edge of the pulse.

6.6.3 Pulse Duration - Pulse duration or pulse width is the time interval in microseconds between the 50 percent of maximum amplitude points on the leading and trailing edges of the pulse.

Source: https://assist.dla.mil -- Downloaded: 202610-17T05:09Z
Check the source to verify that this is the currentversion before use.

6.6.4 Pulse Top - Pulse top is the time interval in microseconds between the 90 percent of maximum amplitude points on the leading and trailing edges of the pulse. Pulse amplitude shall not decay below 80 percent of maximum during this interval.

6.6.5 Pulse Jitter - Pulse jitter is the maximum deviation of the pulse edge from the well defined position of the pulse edge.

* 6.6.6 DC Coupled Circuit - A DC coupled circuit is that portion of a network whose input or output terminal(s) as applicable is (are) conductively connected to ground.

6.6.7 AC Coupled Circuit - An AC coupled circuit is that portion of a network whose input or output terminal(s) as applicable is (are) isolated from the remainder of the network by a series capacitor.

6.6.8 Ripple Voltage - Ripple voltage is the peak-to-peak voltage of the ripple as measured on a calibrated oscilloscope. Measurements shall be made with the receiver gain at minimum and maximum levels.

6.6.9 Useful Video - Useful video is a video output that can be identified on an "A" oscilloscope.

* 6.6.10 Transmitter Operating Frequency - The operating frequency of the transmitter is the frequency of the emitted carrier.

* 6.6.11 Receiver Operating Frequency - The receiver operating frequency (center frequency) is the arithmetic mean between the two frequencies where the overall receiver response is 6 db below maximum.

6.7 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.8 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order.

6.9 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:09Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-55328(WP) RADAR RECOGNITION SET AN/APX-7(*)

ORGANIZATION *(of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426

Source: https://assist.dla.mil -- Downloaded: 20260-17T05:09Z
Check the source to verify that this is the current version before use.

C-6279